

BAHAWALPUR, Pakistan

WMO: 417000

Lat: 29.400N

Lon: 71.783E

Elev: 371

StdP: 14.50

Time Zone: 5.00 (E05)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	43.0	33.7	28.7	51.8	36.1	31.6	52.0	9.9	61.7	8.2	62.8	0.2	N/A	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	19.6	109.4	76.4	107.1	77.1	105.0	77.9	85.1	99.0	84.3	98.2	83.6	97.4	3.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.9	168.2	93.0	80.9	163.0	92.8	80.2	158.7	92.6	50.2	100.2	49.2	98.7	48.3	97.3	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.2	9.4	36.5	113.2	2.1	1.7	35.0	114.4	33.8	115.4	32.7	116.3	31.2	117.6
			35.9	87.2	2.9	0.7	33.8	87.7	32.1	88.1	30.5	88.5	28.3	89.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	55.5	62.1	72.1	84.8	92.5	94.4	93.6	90.8	87.7	80.9	69.7	59.2	(6)
	DBStd	14.42	4.45	5.10	6.76	5.64	4.32	4.15	3.28	2.70	2.99	4.35	4.19	4.90	(7)
	HDD50	14	10	1	0	0	0	0	0	0	0	0	0	3	(8)
	HDD65	614	296	109	16	0	0	0	0	0	0	0	6	187	(9)
	CDD50	10480	179	338	686	1043	1318	1332	1352	1264	1131	957	591	289	(10)
	CDD65	5607	0	27	237	593	853	882	887	799	681	492	148	8	(11)
	CDH74	81086	14	257	2380	7953	13427	14383	14298	12078	9454	5430	1295	117	(12)
	CDH80	51364	0	37	1056	4875	9160	10112	9852	7640	5419	2844	363	6	(13)
(14) Wind	WSAvg	2.4	1.0	1.7	1.7	2.6	3.0	4.6	4.2	4.0	3.0	1.6	0.8	0.7	(14)
(15) Precipitation	PrecAvg	5.1	0.2	0.4	0.4	0.2	0.3	0.3	2.1	0.7	0.6	0.0	0.0	0.1	(15)
	PrecMax	14.0	0.9	3.1	1.7	0.7	1.9	1.5	10.2	3.2	2.6	0.3	0.6	1.7	(16)
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	3.4	0.3	0.8	0.6	0.2	0.5	0.6	2.6	0.9	0.9	0.1	0.2	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.0	84.2	98.5	109.4	112.6	112.3	107.8	103.3	102.1	98.9	89.7	80.7	(19)
		MCWB	60.5	64.2	72.2	71.7	76.0	78.6	82.0	82.8	80.8	76.6	68.2	64.7	(20)
	2%	DB	72.6	80.2	94.3	106.0	109.7	109.2	105.3	101.2	100.1	97.0	86.8	77.2	(21)
		MCWB	58.9	65.0	71.5	71.1	75.5	78.7	82.7	82.1	80.8	75.5	68.5	62.0	(22)
	5%	DB	69.7	77.1	90.0	102.8	107.5	106.8	103.6	99.9	98.3	95.0	84.1	74.4	(23)
		MCWB	56.8	63.5	69.8	71.1	75.4	79.6	82.6	82.0	80.1	74.2	67.3	60.0	(24)
	10%	DB	67.0	73.9	85.9	99.1	105.0	104.4	101.8	98.4	96.7	92.9	81.5	71.6	(25)
		MCWB	55.8	61.3	68.7	70.6	75.3	80.2	82.6	82.2	79.5	73.4	66.0	58.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.5	70.1	75.3	76.4	83.1	85.5	86.3	85.5	85.2	80.8	72.7	65.9	(27)
		MCDB	73.3	78.9	94.0	98.7	102.4	103.5	100.4	97.2	97.6	95.3	84.4	78.0	(28)
	2%	WB	60.2	67.4	73.1	74.9	80.8	84.3	85.2	84.7	83.6	79.1	70.7	63.6	(29)
		MCDB	70.3	77.2	89.5	98.1	102.4	101.9	99.4	96.5	95.8	92.8	83.1	75.2	(30)
	5%	WB	58.4	64.9	71.1	73.5	79.3	83.3	84.4	83.9	82.5	77.2	69.0	61.5	(31)
		MCDB	67.6	74.7	87.0	95.9	100.4	100.8	98.8	95.7	94.3	90.2	81.3	72.3	(32)
	10%	WB	56.7	62.8	69.3	72.1	77.7	82.3	83.7	83.3	81.5	75.5	67.2	59.5	(33)
		MCDB	65.0	71.8	84.1	94.3	99.1	99.7	98.0	94.9	92.9	88.3	78.9	69.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	22.0	22.7	26.0	24.4	19.6	16.2	15.2	17.8	23.5	24.2	23.5	(35)
		MCDBR	26.1	25.7	27.5	30.0	27.0	22.9	17.8	16.7	19.8	23.9	26.1	26.9	(36)
	5% WB	MCWBR	14.3	12.9	10.8	9.2	8.7	6.9	5.2	5.4	6.5	8.2	11.7	13.9	(37)
		MCWBR	23.9	22.8	25.8	26.9	23.2	18.9	16.7	15.6	16.8	21.1	23.7	25.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.494	0.489	0.518	0.597	0.667	0.763	0.810	0.711	0.606	0.605	0.591	0.542	(40)
		taud	1.824	1.888	1.844	1.705	1.601	1.489	1.445	1.606	1.756	1.666	1.653	1.734	(41)
	Ebn at Noon		224	242	246	231	216	195	186	203	220	206	193	200	(42)
		Edh at Noon	56	58	64	76	85	95	98	83	69	71	67	60	(43)
(44) All-Sky Solar Radiation	RadAvg		1041	1384	1784	2080	2218	2122	1948	1882	1831	1576	1198	1025	(44)
	RadStd		90	88	80	67	58	85	94	80	85	68	72	49	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page